Math 133: Chain Rule and Quotient Rule

Name: _____ October 19, 2026

Show your work. Work where each function is differentiable. You may leave answers in factored form.

1. Compute each derivative. Use the differentiation rules you need.

(a) $f(x) = (3x - 2)^5$

(b) $u(x) = \dfrac{2x + 1}{x - 3}$

(c) $h(x) = \sqrt{5x + 4}$

(d) $v(t) = \dfrac{t^2 + 1}{t + 2}$

(e) $g(t) = (t^2 + 4t + 1)^3$

(f) $p(z) = \dfrac{1}{(2z^2 + 3)^2}$

Math 133: Chain Rule and Quotient Rule (continued)

1. (continued)

(g) $r(x) = (x^3 - 2x + 1)^4$

(h) $m(x) = \sqrt{2x+1}\,(x^2+1)$

(i) $q(z) = \dfrac{\sqrt{z+3}}{z+1}$

(j) $s(t) = \sqrt{1 + (t^2+1)^3}$

2. Find an equation of the tangent line to the graph of

$$F(x) = \frac{(x^2+1)^2}{x+1}$$

at $x = 1$. Show your work.